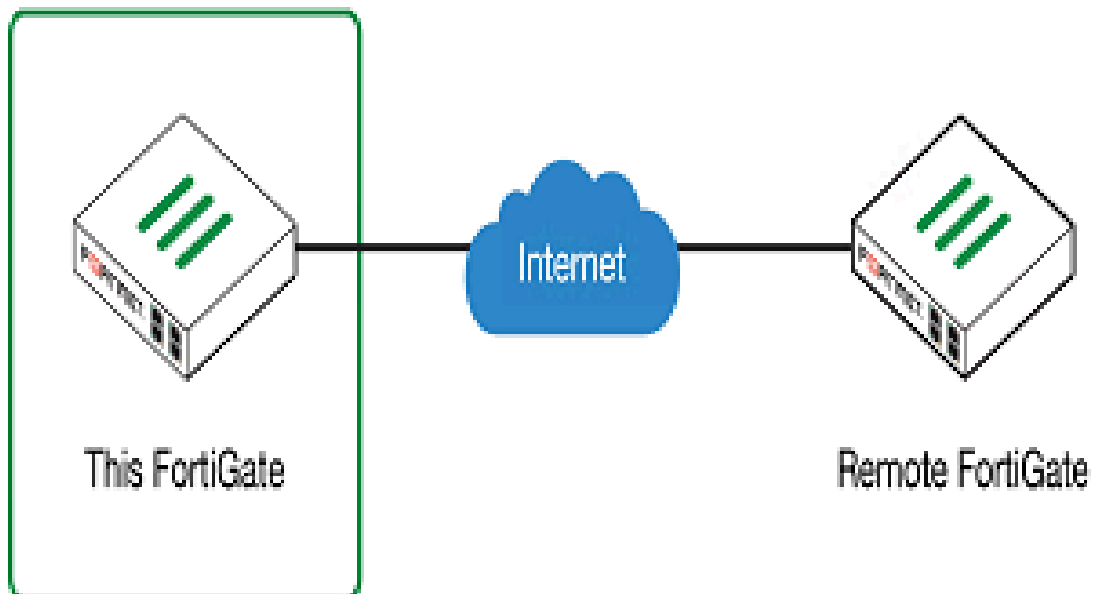


Documentation For IPSEC (site to site)

Site to Site - FortiGate



• VPN >>>> IPsec Wizard

FortiGate-1

VPN Creation Wizard

1 VPN Setup 2 Authentication 3 Policy & Routing 4 Review Settings

Name: HQ-To-BR

Template type: Site to Site

NAT configuration: No NAT between sites

Remote device type: FortiGate

Site to Site - FortiGate

This FortiGate Remote FortiGate

< Back Next > Cancel

NO NAT >>>

Because the purpose of a site-to-site VPN is to make the two networks communicate as if they are one network, using their real internal IP addresses.

So devices in Site A can reach devices in Site B without changing (NATing) any IPs.

FortiGate-1

VPN Creation Wizard

1 VPN Setup 2 Authentication 3 Policy & Routing 4 Review Settings

Remote device: IP Address

Remote IP address: 192.168.1.27

Outgoing Interface: port4

Authentication method: Pre-shared Key

Pre-shared key: Fortinet

Site to Site - FortiGate

This FortiGate Remote FortiGate

< Back Next > Cancel

>>>> Remote IP address :
(IP Address for Remote Forti(BR))

FortiGate-1

VPN Creation Wizard

1 VPN Setup 2 Authentication 3 Policy & Routing 4 Review Settings

Local interface: Users_Vlans

Local subnets:

- 192.168.10.0/24
- 192.168.20.0/24
- 192.168.30.0/24
- 192.168.40.0/24
- 192.168.50.0/24
- 192.168.60.0/24

Remote subnets:

- 172.16.10.0/24
- 172.16.20.0/24
- 172.16.30.0/24
- 172.16.40.0/24
- 172.16.50.0/24
- 172.16.60.0/24

Internet Access: None

Site to Site - FortiGate

This FortiGate Remote FortiGate

< Back Next > Cancel

Local >>> Vlans in HQ

Remote >>> Vlans in BR

Policies Created From VPN to LAN and From LAN to VPN
 >>> to Allow Traffic (@ HQ-forti)

• **Policy & Objects >>> Firewall Policy**

FortiGate-1

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Create New

Edit

Delete

Policy Lookup

Search

Export

Interface Pair View

By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
vpn_HQ-To-BR_remote_0 HQ-To-BR_remote HQ-To-BR_local always ALL ACCEPT Disabled no-inspection All 0B									
Users_Vlans vpn_HQ-To-BR_local_0 HQ-To-BR_local HQ-To-BR_remote always ALL ACCEPT Disabled no-inspection All 0B									
Users_Vlans port1									
Users_Vlans Server_Vlan60									
Users_Vlans Users_Vlans									

Edit Policy

Name

vpn_HQ-To-BR_local_0

Incoming Interface

Users_Vlans

Outgoing Interface

HQ-To-BR

Source

HQ-To-BR_local

+

Destination

HQ-To-BR_remote

+

Schedule

always

Service

ALL

+

Action

ACCEPT

DENY

Inspection Mode

Flow-based

Proxy-based

Firewall / Network Options

NAT
 ☐

Edit Policy

Name

vpn_HQ-To-BR_remote_0

Incoming Interface

HQ-To-BR

Outgoing Interface

Users_Vlans

Source

HQ-To-BR_remote

+

Destination

HQ-To-BR_local

+

Schedule

always

Service

ALL

+

Action

ACCEPT

DENY

Inspection Mode

Flow-based

Proxy-based

Firewall / Network Options

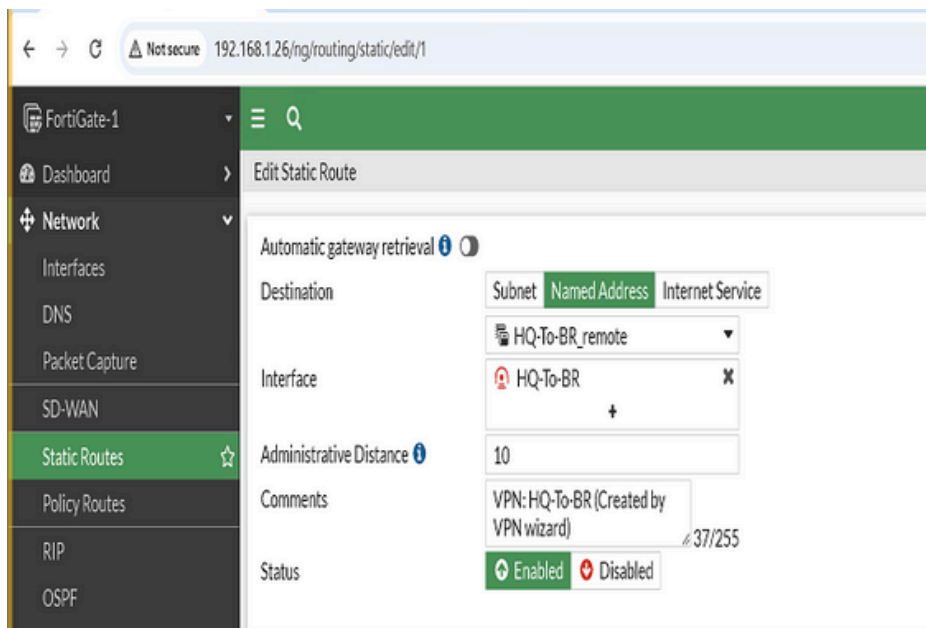
NAT
 ☐

Protocol Options

PROT

default

- **Network >>> Static Route**



Because the firewall needs to know:

“If I want to reach the remote network... which path should I take?”

The FortiGate must be told:

If the traffic is going to 172.16.10.0/24 (the remote site’s network)

→ send it through the IPsec Tunnel interface.

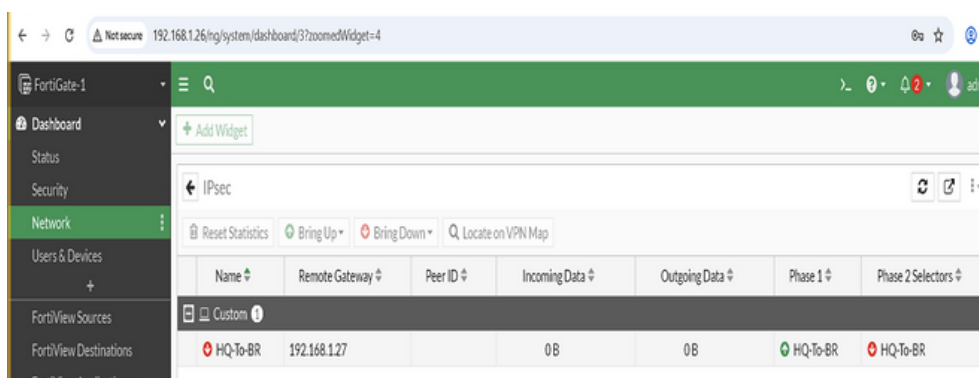
Without the static route:

The device knows the VPN tunnel exists

But it doesn’t know that it should send the traffic inside the tunnel

So ping and communication will fail.

- **Dashboard >>> Network >>> IPsec**

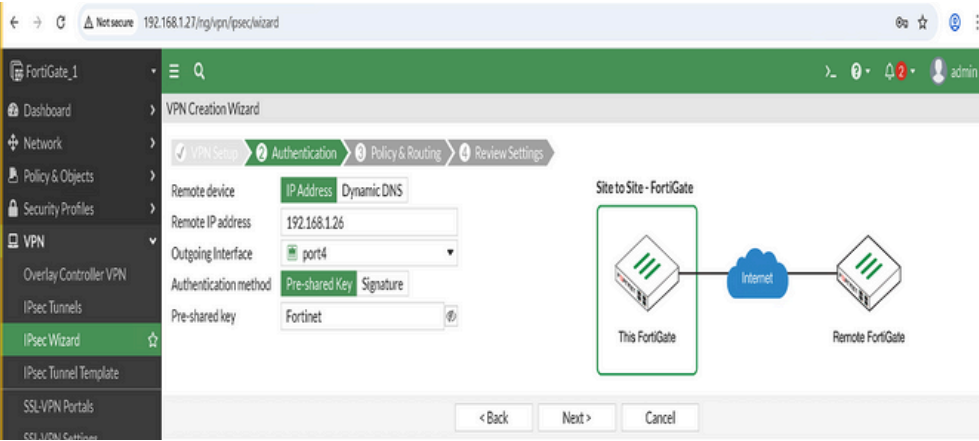
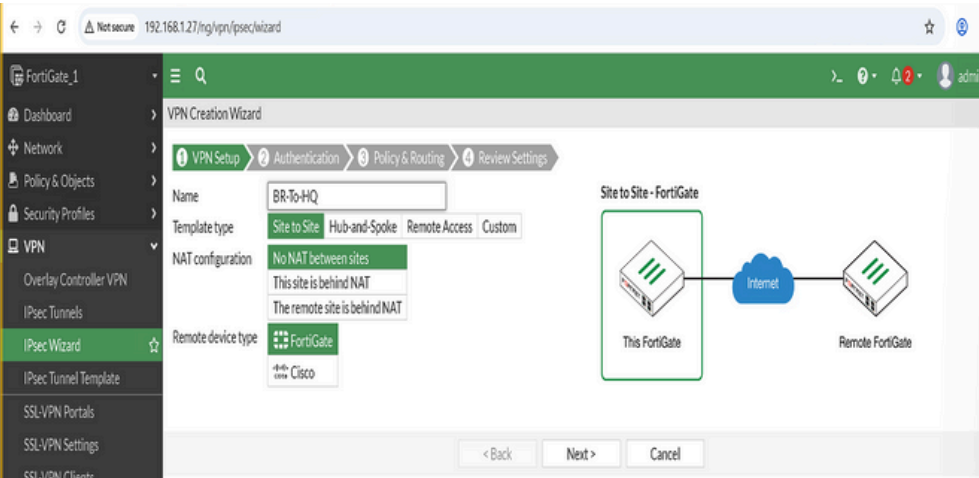


After we show tunnel and custom it

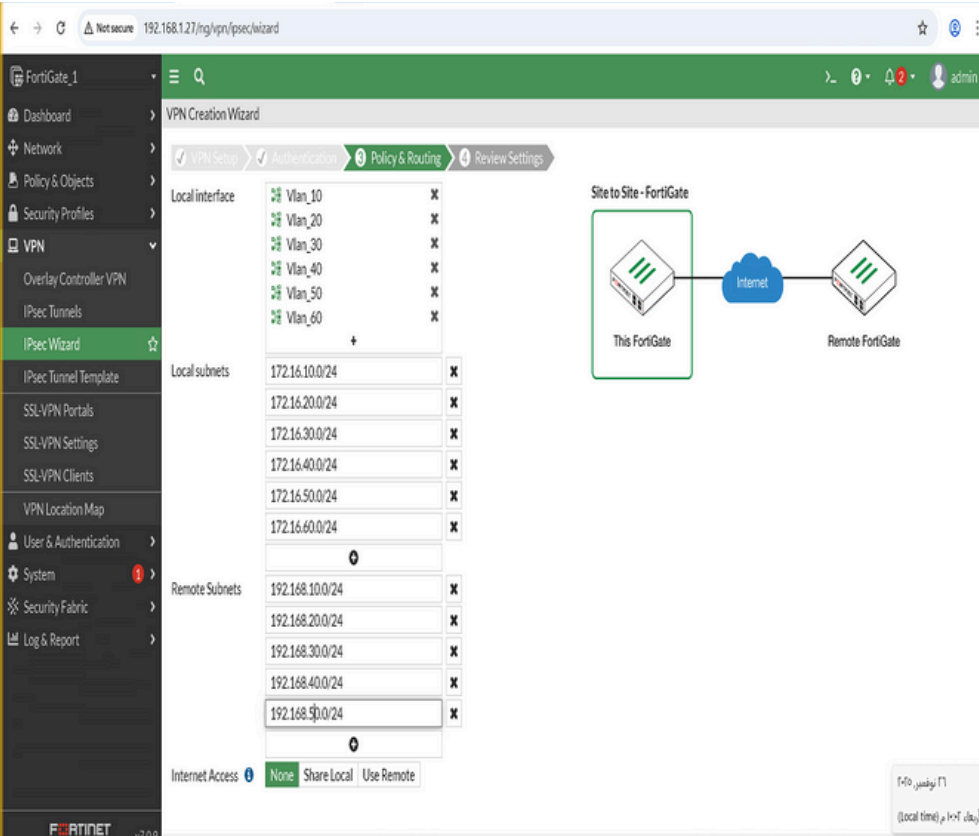
>>> We need to Make sure of status of tunnel and show traffic

Repeat the steps for the second forti (BR)

• VPN >>>> IPsec Wizard



We must use the same
Pre-shared Key
[Fortinet]



Local >>> Vlans in BR

Remote >>> Vlans in HQ

- **Policy & Objects >>> Firewall Policy**

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
BR-To-HQ → MGT (port1)									
vpn_BR-To-HQ_remote0	BR-To-HQ_remote	BR-To-HQ_local	always	ALL	ACCEPT	Disabled	no-inspection	All	0B
MGT (port1) → BR-To-HQ									
vpn_BR-To-HQ_local_0	BR-To-HQ_local	BR-To-HQ_remote	always	ALL	ACCEPT	Disabled	no-inspection	All	0B

Automatically Created But we Can modify it if number of policies has exceeded the allowed limit.

Edit Policy

Name

vpn_BR-To-HQ_local_0

Incoming Interface

MGT (port1)

Outgoing Interface

BR-To-HQ

Source

BR-To-HQ_local

Destination

BR-To-HQ_remote

Schedule

always

Service

ALL

Action

ACCEPT

DENY

Inspection Mode

Flow-based

Proxy-based

Firewall / Network Options

NAT

Protocol Options

default

Security Profiles

Edit Policy

Name

vpn_BR-To-HQ_remote0

Incoming Interface

BR-To-HQ

Outgoing Interface

MGT (port1)

Source

BR-To-HQ_remote

Destination

BR-To-HQ_local

Schedule

always

Service

ALL

Action

ACCEPT

DENY

Inspection Mode

Flow-based

Proxy-based

Firewall / Network Options

NAT

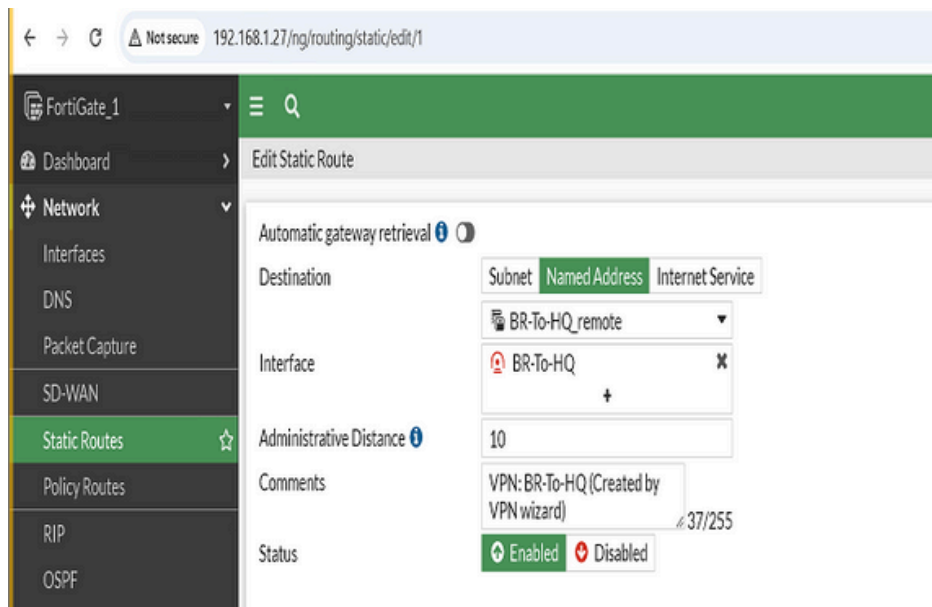
Protocol Options

default

Security Profiles

Outgoing Interface >> Port 1 (users)

- **Network >>> Static Route**



The next step is to confirm that both Phase 1 and Phase 2 parameters are fully aligned on both FortiGate devices.

- **VPN >>>> IPsec Tunnel**

After Custom the tunnel we can edit here

HQ

Remote Gateway: Static IP Address (192.168.1.27), Interface: port4

Authentication

Method: Pre-shared Key

Pre-shared Key:

IKE

Version: 1 2

Mode: Aggressive Main (ID protection)

Phase 1 Proposal

Algorithms: DES-MD5, DES-SHA256

Diffie-Hellman Groups: 14, 5

BR

Remote Gateway: Static IP Address (192.168.1.26), Interface: port4

Authentication

Method: Pre-shared Key

Pre-shared Key:

IKE

Version: 1 2

Mode: Aggressive Main (ID protection)

Phase 1 Proposal

Algorithms: DES-MD5, DES-SHA256

Diffie-Hellman Groups: 14, 5

Verify the following parameters on both sides of the tunnel:

Authentication method: **Pre-Shared Key**

IKE Version: **IKEv1**

Exchange Mode: **Main Mode** (chosen because it provides **better security** by protecting the identities during negotiation even if it is slower than Aggressive Mode)

Encryption: **DES**

Integrity (Hashing): **MD5** and **SHA-256**

Next, verify that the **Phase 2** settings match on both sides of the tunnel.

The following parameters must be identical:

Local Subnet and Remote Subnet (Selectors / Quick Mode Selectors)

Make sure both FortiGates are using the same local and remote networks in Phase 2.

Encryption Algorithm : DES

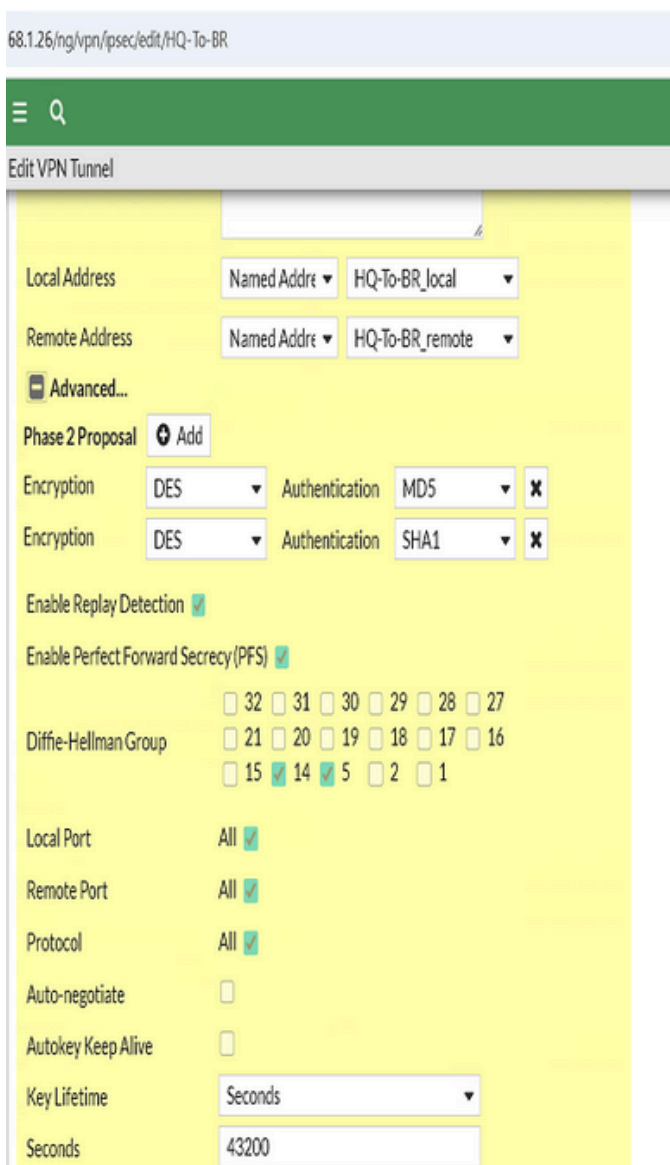
Integrity (Hashing) Algorithm: MD5 , SHA1 or SHA-256

PFS (Perfect Forward Secrecy)

Must be the same on both sides — either enabled with the same DH group or disabled.

Lifetime: Both devices must use the same Phase 2 lifetime (43200 seconds).

HQ



68.1.26/ng/vpn/ipsec/edit/HQ-To-BR

Edit VPN Tunnel

Local Address: Named Addr ▾ HQ-To-BR_local ▾

Remote Address: Named Addr ▾ HQ-To-BR_remote ▾

Advanced...

Phase 2 Proposal: Add

Encryption: DES ▾ Authentication: MD5 ▾ X

Encryption: DES ▾ Authentication: SHA1 ▾ X

Enable Replay Detection ☒

Enable Perfect Forward Secrecy(PFS) ☒

Diffie-Hellman Group: ☐ 32 ☐ 31 ☐ 30 ☐ 29 ☐ 28 ☐ 27 ☐ 21 ☐ 20 ☐ 19 ☐ 18 ☐ 17 ☐ 16 ☐ 15 ☒ 14 ☒ 5 ☐ 2 ☐ 1

Local Port: All ☒

Remote Port: All ☒

Protocol: All ☒

Auto-negotiate: ☐

Autokey Keep Alive: ☐

Key Lifetime: Seconds ▾

Seconds: 43200

BR



68.1.27/ng/vpn/ipsec/edit/BR-To-HQ

Edit VPN Tunnel

Local Address: Named Addr ▾ BR-To-HQ_local ▾

Remote Address: Named Addr ▾ BR-To-HQ_remote ▾

Advanced...

Phase 2 Proposal: Add

Encryption: DES ▾ Authentication: MD5 ▾ X

Encryption: DES ▾ Authentication: SHA1 ▾ X

Enable Replay Detection ☒

Enable Perfect Forward Secrecy(PFS) ☒

Diffie-Hellman Group: ☐ 32 ☐ 31 ☐ 30 ☐ 29 ☐ 28 ☐ 27 ☐ 21 ☐ 20 ☐ 19 ☐ 18 ☐ 17 ☐ 16 ☐ 15 ☒ 14 ☒ 5 ☐ 2 ☐ 1

Local Port: All ☒

Remote Port: All ☒

Protocol: All ☒

Auto-negotiate: ☐

Autokey Keep Alive: ☐

Key Lifetime: Seconds ▾

Seconds: 43200

We need to confirm that both **FortiGate devices can see each other** over the VPN tunnel.

This ensures that Phase 1 and Phase 2 negotiations have successfully completed, and the tunnel is established and ready to **carry traffic**.

- *execute ping <ip for remote fortigate >*

```
CLI Console(1)
eFortiGate-1 # e
Unknown action 0

FortiGate-1 # execute ping 192.168.1.27
PING 192.168.1.27 (192.168.1.27): 56 data bytes
64 bytes from 192.168.1.27: icmp_seq=0 ttl=255 time=25.4 ms
64 bytes from 192.168.1.27: icmp_seq=1 ttl=255 time=3.5 ms
64 bytes from 192.168.1.27: icmp_seq=2 ttl=255 time=6.8 ms
64 bytes from 192.168.1.27: icmp_seq=3 ttl=255 time=6.9 ms
64 bytes from 192.168.1.27: icmp_seq=4 ttl=255 time=7.4 ms

--- 192.168.1.27 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 3.5/10.0/25.4 ms

FortiGate-1 #
```

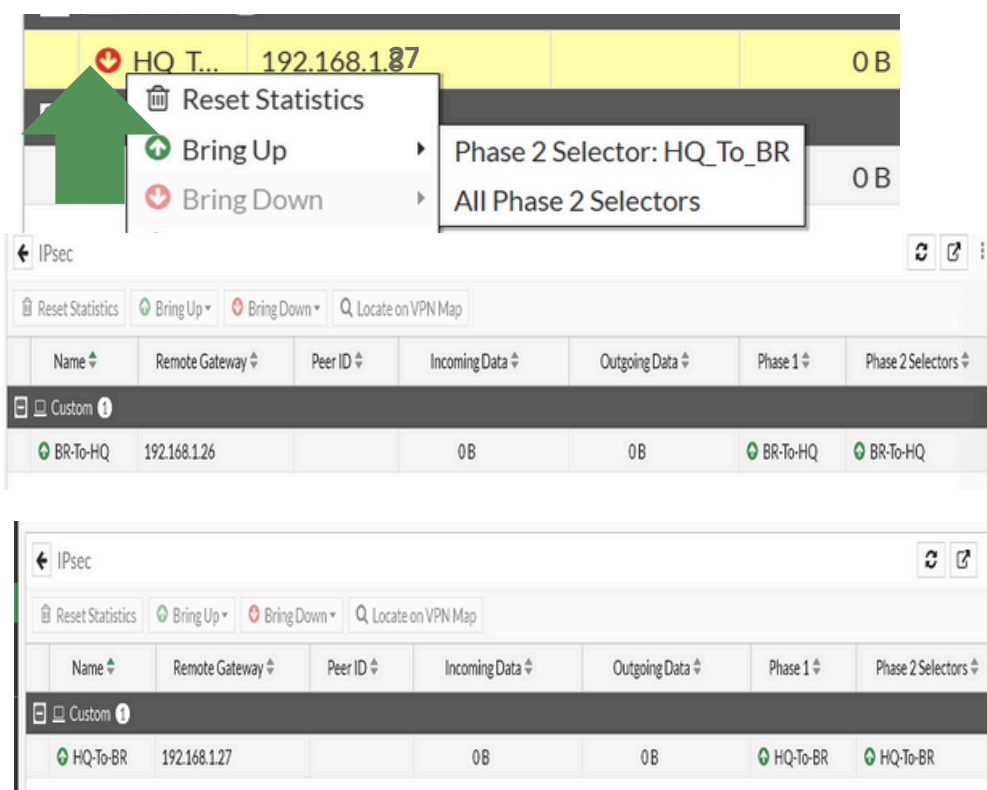
```
CLI Console(1)
FortiGate_1 # execute ping 192.168.1.26
PING 192.168.1.26 (192.168.1.26): 56 data bytes
64 bytes from 192.168.1.26: icmp_seq=0 ttl=255 time=59.1 ms
64 bytes from 192.168.1.26: icmp_seq=1 ttl=255 time=6.6 ms
64 bytes from 192.168.1.26: icmp_seq=2 ttl=255 time=6.8 ms
64 bytes from 192.168.1.26: icmp_seq=3 ttl=255 time=7.3 ms
64 bytes from 192.168.1.26: icmp_seq=4 ttl=255 time=6.7 ms

--- 192.168.1.26 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 6.6/17.3/59.1 ms

FortiGate_1 #
```

To verify that the tunnel is up and ready on both sides

- *Dashboard >>> Network >>> IPsec*



- *Right click >> bring up >> all phase 2 selectors*

Both Phase 1 and Phase 2 are UP, and the IPsec tunnel is currently ready to carry traffic between the two sites.